MILITARY SPECIFICATION

CHARGE, PROPELLANT, INCREMENT, XM185 (MADE FROM PROPELLANT, M9)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Charge. Propellant, Increment, XM185, made from M9 Propellant for use in Cartridge, 81MM, Illuminating, XM652.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Flat, Steel

MILITARY

MIL-B-2427	- Boxes, Ammunition Packing; Wood, Nailed
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-B-3106	- Board, Compositions, Water-Resistant, Solid
MIL-L-10547	- Liners, Case, Waterproof
MIL-B-13239	- Barrier Material, Waterproofed, Flexible, All Temperature
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1375

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MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8796522 - Marking Diagram and Sealing for Wooden Packing Boxes
20-4-619 - Marking Diagram and Sealing of Fiberboard Cartons for Shipment of Propellant Increment
76-4-59 - Box, Packing, Metal-Wood, M18 for Smokeless Powder Assembly
76-4-46 - Box, Packing with Metal Liner, M24 for Smokeless Powder - Assembly and Details
76-4-56 - Box, Packing, Metal-Wood, M17, for Smokeless Powder
7548645 - Carton, Packing, Reusable Collapsible for High Explosives, Assembly Details, Packing and Marking
7549033 - Container, Metal, Universal, M25, for Propellants and Explosives
8858577 - Marking Diagram and Sealing of Container, Metal, Universal, M25, for Shipment of Propellants
8858848 - Marking, Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellants
9220706 - Charge, Propellant, Increment, XM185

PUBLICATIONS

ARMY

ET-9220706 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.-The following document forms a part of this specification to the extent specified herin. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 71-90 (latest revision)".)

3 REQUIREMENTS

3.1 Material - Material shall be in accordance with applicable drawings and specifications.

3.2 Charge.-The charge shall comply with all requirements specified on Drawing (dwg.) 9220706) and all requirements specified in the applicable drawings and specifications.

3.3 Workmanship.-The propellant increment charge shall show no visible evidence of moisture and shall be free of tears, rips and foreign matter. The stitches on the bag shall be neat and continuous without breaks, tears,or foreign matter. All required marking shall be neat and clearly defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The

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provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an indentification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the-contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an

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essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot of increments shall be prepared from M9 propellant composition from batches containing the same proportion of ingredients and displaying uniformity in grain thickness.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bag, Propellant (see dwg. 9220707 covering a detail of dwg. 9220706).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major: None defined.

Minor:		AQL 4.0 percent	
201.	Width	Gage. . . .	01001
202.	Length of button hole (2 places) minimum.....	Gage	01002
203.	Location of button hole (2 places).....	Gage	01003
204.	Total length.....	Gage	01004
205.	Location of stitching....	Visual-Scale	01005
206.	Incorrect number of stitches per inch.....	Visual-Scale	01006
207.	Stitches broken or missing.....	Visual	01007
208.	Marking missing, misleading or unidentifiable.....	Visual	01008
209.	Evidence of poor workmanship (see 3.3).....	Visual	01009

4.2.2.2 Charge, Propellant, Increment (see dwg. 9220706).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Propellant exposed or escaping from bag.....	Visual	02001

Major: None defined.

Minor:		AQL 1.5 percent	
201.	Incorrect number of transverse lock stitches.....	Visual-Scale	02002
202.	Transverse thread ends not extended.....	Visual	02003
203.	Evidence of poor workmanship (see 3.3).....	Visual	02004
204.	Marking misleading or unidentifiable.....	Visual	02005

4.2.2.3 Sealed inner liner (see 5.1.1, 5.1.2 or 5.1.3).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major:	AQL .40 percent		
101.	Liner damaged so that contents are exposed or liable to become exposed.....	Visual	03001
102.	Bag incompletely sealed.....	Visual	03002

Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.3).....	Visual	03003

4.2.2.4 Sealed Wood Packing Box (see 5.1.1, 5.1.2 or 5.1.3).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken or loose.....	Visual-Manual	04001
102.	Board broken or split.....	Visual	04002
103.	Contents exposed.....	Visual	04003
Minor:	AQL 1.5 percent		
201.	Strapping improperly engaged....	Visual-Manual	04004
202.	Contents move when shaken.....	Manual	04005
203.	Marking misleading or unidentifiable.....	Visual	04006

4.2.3 Testing

4.2.3.1 Weight of propellant (see dwg. 9220706) - Major defect - Code No. 02006.-The propellant charge shall be weighed 100 percent and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weight determination, and if performed manually shall be performed by a different operator. Any propellant charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.2 Air pressure test (see dwg. 8858848) - Major defect - Code No. 03004.-The containers shall be tested 100 percent in accordance with 4.3.1. If any container fails to meet the requirement it shall be rejected and removed from the lot.

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4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-9220706 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Air pressure test for containers (see dwg. 8858848).-The container shall be subjected to an internal pressure of 1/2 to 1 pound per square inch by a method satisfactory to the Government. A water manometer shall be assembled in the system. A drop of 0.7 inches or more on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 PACKING

5.1.1 Level A.-Propellant increments shall be packed in barrier bags complying with Type I, Class E of Specification MIL-B-117. The barrier bag size shall be approximately 29-1/2 inches wide by 22 inches long. Approximately 15 pounds of propellant shall be packed in each bag and closed by means of heat sealing.

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Heat seals shall comply with the heat seal test of MIL-P-116. Two heat sealed bags shall be packed in a wooden box conforming to Grade A, Type I, Class 1 of Specification MIL-B-2427 except that sides, top and bottom may be 1/2 inch thick lumber. Ends, sides, top and bottom of the wooden box shall be lined with composition board conforming to Type I or II of Specification MIL-B-3106. The box shall be closed and strapped in accordance with Appendix of the box specification. The net weight shall not exceed 30 pounds of propellant.

5.1.2 Level B.-Propellant increments shall be packed in propellant packing boxes M17, M18, M24 or M25 complying with dwgs. 76-4-46, 76-4-59, 76-4-56 or 7549033 respectively. Containers shall be closed, sealed and tested in accordance with dwg. 8858848.

5.1.3 Level C.-Propellant increments shall be packed in accordance with dwg. 7548645, Carton Number 1, except that the bag liner shall be manufactured from barrier material conforming to Grade A, Type CW-2, Specification MIL-B-13239.

5.2 Marking

5.2.1 Level A.-Exterior containers shall be marked in accordance with Drawing 8796522. The ICC nomenclature to be applied shall be "Propellant Explosive - Class A".

5.2.2 Level B.-The M17, M18, and M24 containers shall be marked in accordance with Drawing 8858848. The M25 container shall be marked in accordance with Drawing 8858577.

5.2.3 Level C.-The fiberboard carton shall be marked in accordance with Drawing 20-4-619.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when any Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-PA

Preparing activity:
Army-PA

Project Number: 1375-A-1065

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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